PB

PAUL BORGEN

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Personal Summary

Results-driven professional with extensive experience in architecting scalable, resilient systems and leading diverse teams. Proven ability to enhance a team's ability to deliver through AI-driven solutions and innovative project execution, ensuring compliance with quality standards. Strong communicator adept at building relationships with stakeholders and driving strategic initiatives.

Skills

- Software Architecture
- System Design
- Team Leadership
- AWS
- AI
- Claude / Codex / Grok
- LLM's
- DAG
- LangChain / LangGraph
- Mentoring
- CI/CD Pipelines
- Terraform
- Python
- Golang
- TypeScript
- Java
- Bash
- Smart Contract
- Blockchain
- Data Processing
- Computer Vision
- Automated Testing
- Mongo
- Redis
- MYSQL / Postgress / Oracle

Work Experience

Alopex

Principal AI Engineer

05/2024 - Current

- Architected scalable, HIPAA-compliant, multi-tenant patient management systems on AWS using Python, Golang, to power secure care coordination across medical organizations.
- Designed and built an AI-powered clinical chat interface leveraging AWS, LLMs, vector databases, and Directed Acyclic Graphs (DAGs) to provide nurses with real-time, context-rich patient summaries, dramatically improving situational awareness and decision-making speed.
- Architected a scalable AI platform on AWS using LangChain, Amazon Bedrock, and LangGraph agents running on AWS Fargate (ECS). Integrated OpenSearch Serverless for embeddings, S3 for document processing, Redis caching, and Step Functions for orchestration, enabling resilient and complex AI workflows.
- Designed secure authentication, role-based access control (RBAC), and intelligent workflows for referrals, eligibility verification, and care team coordination while maintaining strict data privacy standards.

Mintra

Co-Founder / CTO

07/2021 - 05/2024

- Leading a global distributed team to build all aspects of the NFT marketplace.
- Built an LLM-powered NFT recommendation engine that analyzed user behavior, collection traits, and on-chain activity to deliver personalized NFT suggestions, significantly improving user engagement and discovery on the platform.
- Used RunPod Hosted machines to run Ollama based Models for the recommendation system.
- Leveraged modern AI development tools (including Cursor / Claude) to accelerate software development velocity, enabling faster iteration, code generation, and higher-quality output across the engineering team.
- Ensured all smart contracts underwent a full audit and have proven secure to date, using Foundry and Hardhat with a comprehensive suite of unit tests.
- Architected a data pipeline that has processed millions of NFTs and tens of thousands of collections.
- Developed a distributed locking workflow capable of processing blockchain data in a distributed, deterministic manner.
- Mintra exemplifies innovation and technical expertise, emerging as the leading NFT marketplace on Pulsechain.com.

Precision Point Systems

Software Engineering Consultant

02/2020 - 07/2021

- Engaged to evaluate and refine a sophisticated computer vision system designed for precise tracking of a football on the field.
- This project leveraged advanced C / C++ / GoLang programming to manage an array of high-resolution cameras monitoring the game in real-time.
- Our task involved capturing images from these cameras and employing a custom AI model that I trained to determine the exact position of the football with pinpoint accuracy.
- OpenCV was integral throughout this process.
- This initiative not only showcased cutting-edge technology in sports analytics but also expanded the capabilities of computer vision applications in dynamic environments.
- Stabilized the system for reliable use during live games.
- Successfully deployed the system for UCONN football games at Rentschler Field.
- Achieved measurement accuracy down to the pixel level from the camera, equivalent to approximately half an inch.

Honeywell

Staff Software Engineer (Solutions / DevOps / Analytics)

06/2016 - 02/2020

- Developed and maintained multiple external interfaces within metering infrastructures.
- Executed key operations including database design, performance optimization, and backend development.
- Strengthened security by enhancing the enterprise security API (ESAPI) and reducing vulnerabilities to XSS and XSRF attacks.

- Drove performance improvements through efficient SQL queries and advanced threading strategies.
- Updated several backend applications to utilize Spring and Hibernate, resulting in a simplified architecture and faster turnaround times in subsequent sprints.
- Customized and applied ESAPI implementations to enhance security across all business suite applications.
- Led security reviews by analyzing penetration testing reports and implementing required fixes.
- Introduced unit testing methodologies and strategies to improve code quality within the team.
- Migrated the codebase from SVN to GIT to streamline version control workflows.
- Created a CI pipeline using Jenkins, enabling end-to-end automation in the continuous integration process.
- Led distributed teams and innovated SDLC and communication methods to enhance collaboration and delivery.

Honeywell

Senior Software Engineer

07/2012 - 06/2016

- Developed and maintained multiple external interfaces within metering infrastructures.
- Executed key operations including database design, performance optimization, and backend development.
- Enhanced security by advancing the enterprise security API (ESAPI) and reducing vulnerabilities to XSS and XSRF attacks.
- Drove performance improvements through efficient SQL queries and threading strategies.
- Noted Accomplishments: Updated several backend applications to leverage Spring and Hibernate, resulting in simpler architecture and faster sprint turnaround times.
- Customized and utilized ESAPI implementation to strengthen security across all applications in the business suite.
- Introduced and established unit testing methodologies and strategies within the team.
- Migrated the codebase from SVN to GIT.
- Created a CI pipeline with Jenkins and enabled end-to-end automation in the CI process.
- Led offshore teams over multiple years within a Scrum SDLC environment.

CARQUEST Technologies, LLC

Web Application Developer

06/2010 - 07/2012

- Worked on innovating the auto parts inventory and management system by analyzing existing VB6 applications to understand data flow.
- Migrated and integrated this data into a J2EE application I developed, utilizing a combination of Java and PLSQL to enhance data processing speed.

Pratt & Whitney

QA Team Leader

08/2008 - 06/2010

- Spearheaded the automation of testing processes for applications processing real-time jet engine data via FADEC (Full Authority Digital Engine Control).
- Collaborated closely with aerospace engineers to gather and refine testing requirements, ensuring alignment with engineering objectives and operational constraints.
- Translated engineering requirements into detailed test plans, leading the team through manual testing to validate application performance.
- Designed and implemented comprehensive automation suites by converting manual test plans into automated tests, significantly enhancing efficiency and reducing error rates.
- Managed team workflows, delegated tasks effectively, and facilitated seamless collaboration between testers and engineers to deliver robust, reliable testing solutions.
- Maintained a testing suite in Mercury Quality Center.
- Validated 4 J2EE applications.
- Developed detailed Test Plans.
- Created a Java program to verify database population accuracy.
- Collaborated with top Aerospace and Mechanical engineers to define testing requirements.
- Built testing suites using VBScript (QTP) and later Selenium for automated functional testing.

Education

Central Connecticut State University

Bachelor of Science - BS in Computer Science

Websites, Portfolios, Profiles

- https://github.com/pborgen
- https://www.paulborgen.com
- https://github.com/pborgen

Certifications

- AWS Certified Cloud Practitioner
- AWS Certified Solutions Architect - Associate
- Certified Ethical Hacker, EC-Council, http://www.eccouncil.org/Certification/certified-ethical-hacker

Hobbies and Interests

- Gym
- Pickleball
- Poker